A kinetic-impact deflection model

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State the simplified collision model.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to explain your model or result. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

impulse: net force acting over time; change in momentum

01 Warm-up: bring back an earlier idea

Two equal 3 kg carts approach at 5 m/s in opposite directions and stick.
Find their common final velocity and explain.

Word help: final = at the end of the interval; subscript f

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × time.
Toy model: much heavier target; straight-line shift.

Worked reasoning: In a fictional model, a 2 kg projectile at 30 m/s sticks to a stationary 10 kg target. Final speed is 60/12=5 m/s. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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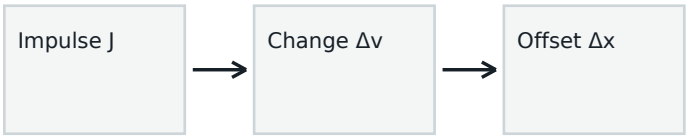
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

After a brief sideways push, a probe has a constant additional sideways velocity.
Compare the offset after a short drift and a long drift.
Use the model to connect impulse, velocity change and later position. Assume no further sideways force.

Sentence starter: Both ... . They differ in ... because ... .

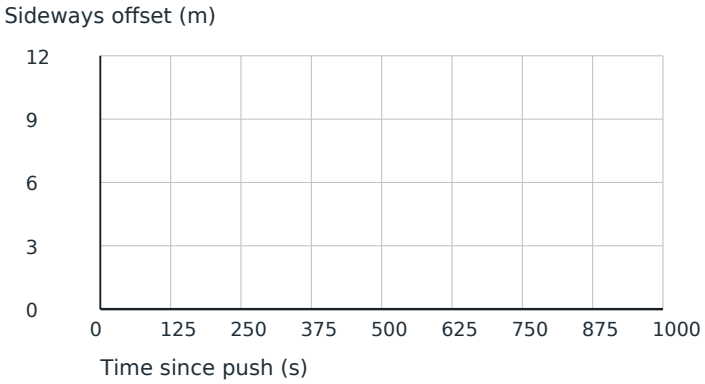


Marzano check - circle: ① ② ③

04 Apply the physics

A simulated probe receives a brief push that adds 0.01 m/s sideways to its velocity. It then drifts without further sideways force. Complete the graph and find its sideways offset after 100 s and 1000 s.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An asteroid's period is 6 years.
Explain why that number alone cannot establish an Earth encounter.
Name two additional observations needed.
Organize the timing evidence before drawing a conclusion.

Word help: encounter = objects reaching nearby positions at the same time

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Object	Arrival time / window	Comparison
A	____	____
B	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

At the same small Δv, compare shift with twice as much lead time.
State one model limit. Organize the known quantities on the before-and-after model.

Sentence starter: Both … . They differ in … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Keep projectile momentum 60 kg·m/s but double the stationary target mass from 10 to 20 kg.
Calculate the new final speed and explain the change. Organize the known quantities on the before-and-after model.

Sentence starter: My claim is … . The diagram/data show …, so … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3